▸ SANTI SPADARO, *The cellular-Lindelöf property and cardinal invariants of (and above) the continuum.*
Department of Mathematics and Computer Science, University of Catania, viale Andrea Doria n. 6, 95125 Catania, Italy.
E-mail: `sspadaro@unict.it`.

A topological space X is said to be *cellular-Lindelöf* [2] if for every family of pairwise disjoint non-empty open subsets of X (henceforth called a *cellular family*) there is a Lindelöf subspace L of X which meets every member of the family. A space is called *almost cellular-Lindelöf* [1] if for every cardinal κ and for every cellular family $\mathcal{U}$ in X such that $|\mathcal{U}| = \kappa$ there is a Lindelöf subspace L of X which meets κ many members of $\mathcal{U}$.

Cellular-Lindelöf spaces are a common generalization of Lindelöf spaces and spaces with the countable chain condition, that was originally motivated by the problem of finding a common extension to Arhangel'skii's Theorem and the Hajnal-Juhász inequality (see [2]). While solving that problem required a shift in perspective (see [4]), the question of whether every cellular-Lindelöf first-countable regular space has cardinality at most continuum is still open, and various partial answers to it have appeared in the literature (see, for example, [3], [7] and [8]).

After giving an introduction to cellular-Lindelöf spaces, we shall present three new examples regarding this class of spaces, the first two of which solve questions of Ofelia Alas, Luis Enrique Gutiérrez-Dominguez and Richard Wilson from [1]. Their construction involves the classical cardinal invariants of the continuum (or their stepped-up versions). Namely, we will construct:

1. A consistent example (from $\mathfrak{t}_{\omega_1} = \omega_2$) of a normal almost cellular-Lindelöf space which is neither cellular-Lindelöf nor weakly Lindelöf.
2. A consistent example (from $\mathfrak{b} = \omega_1$) of a cellular-Lindelöf Tychonoff space with uncountable weak Lindelöf degree for closed sets whose Lindelöf degree doesn't exceed ω_1.
3. A ZFC example of a space whose cellular-Lindelöf property is independent of ZFC (and whose normality also turns out to be independent of ZFC).

Our talk is based on joint work with Rodrigo Hernández-Gutiérrez (UAM, Mexico City) from [6].

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[2] A. Bella and S. Spadaro, *On the cardinality of almost discretely Lindelöf spaces*, Monatshefte für Mathematik vol. 186 (2018), 345–353.

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[4] A. Bella and S. Spadaro, *A common extension of Arhangel'skii's Theorem and the Hajnal-Juhász's inequality*, Canadian Mathematical Bulletin vol. 63 (2020), 197–203.

[5] A. Dow and R.M. Stephenson, *Productivity of cellular-Lindelöf spaces*, Topology and its Applications vol. 290 (2021), Article ID 107606.

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[7] I. Juhász, L. Soukup and Z. Szentmiklóssy, *On cellular-compact spaces*, Acta Mathematica Hungarica vol. 162 (2020), 549–556.

[8] V.V. Tkachuk and R.G. Wilson, *Cellular-compact spaces and their applications*,

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